TASTY BYTES

OPTIMIZING THE DISPLAY OF RECIPES ON THE HOMEPAGE
FOR GENERATING HIGH USER TRAFFIC
AND
ACHIEVING BUSINESS SUCCESS

CONTENT

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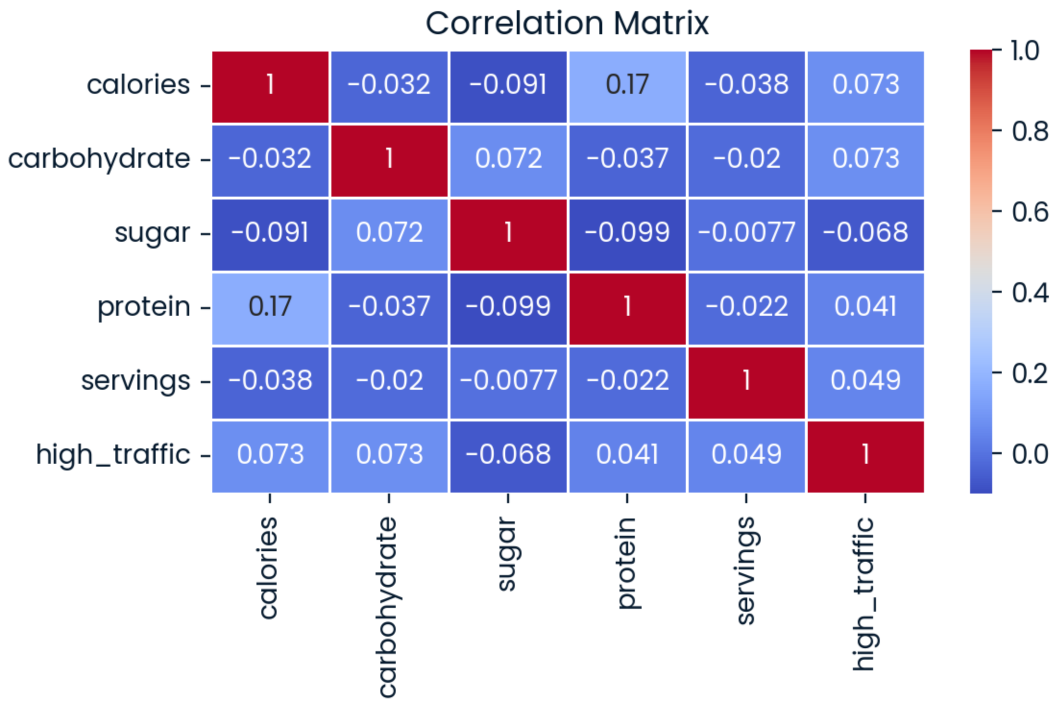
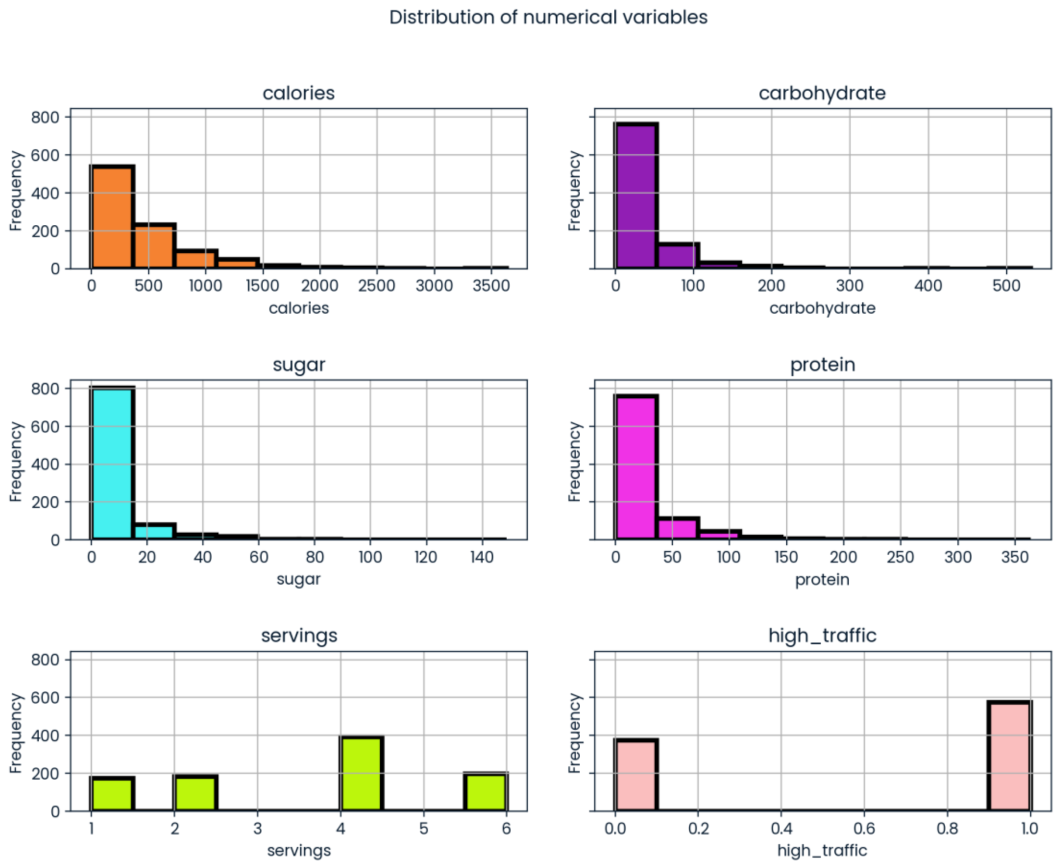
OVERVIEW OF THE PROJECT

- This project focused on identifying and **predicting high-traffic recipes** to optimize homepage content and with this drive increased user engagement and subscriptions.
- By applying machine learning **models** and thorough data analysis, the project aimed to ensure that popular recipes are featured prominently, maximizing potential user interest.
- **Binary Classification** challenge with a target variable: high-traffic or not



BUSINESS GOALS

- Supporting data-driven decisions for recipe selection
- Predict recipes leading to high website traffic.
- Correctly predict high traffic recipes 80% of the time.
- Increase Web Traffic and User Engagement
- Subscription Growth



DATA SCIENCE SOLUTION

- **Data Cleaning:**
  - Initial data validation and cleaning ensured a high-quality dataset by handling missing values and inconsistencies.
- **Data Preparation**:
  - Feature Engineering, variable encoding, numerical scaling, and SMOTE (Synthetic Minority Over-sampling Technique) balanced the dataset, addressing class imbalance in high-traffic recipes.
- **Data Exploration:**
  - A diverse range of statistical methods and test were applied, including correlations and t-tests.
  - Visual insights were provided using bar plots, histograms, and ROC curves.
  - Main question: “Is the relationship between features and target variable linear or not?
  - **Result**: the relationship is not linear and non-linear binary ML-models are a considered a better choice.
- **Model Development and Tuning:**
- Various classification models: logistic regression, decision trees, SVM, gradient boosting, and XGBoost
  Those were trained and tuned.
- **Evaluation:**
  - The models were evaluated on **Recall**.
- **Model Selection**
  - The final model was the hyper tuned Decision Tree Classifier with a recall over 80 percent and the best Receiver Operating Curve with an AUC around 70 percent.

DATA SCIENCE EVALUATION METRIC: RECALL

Definition:

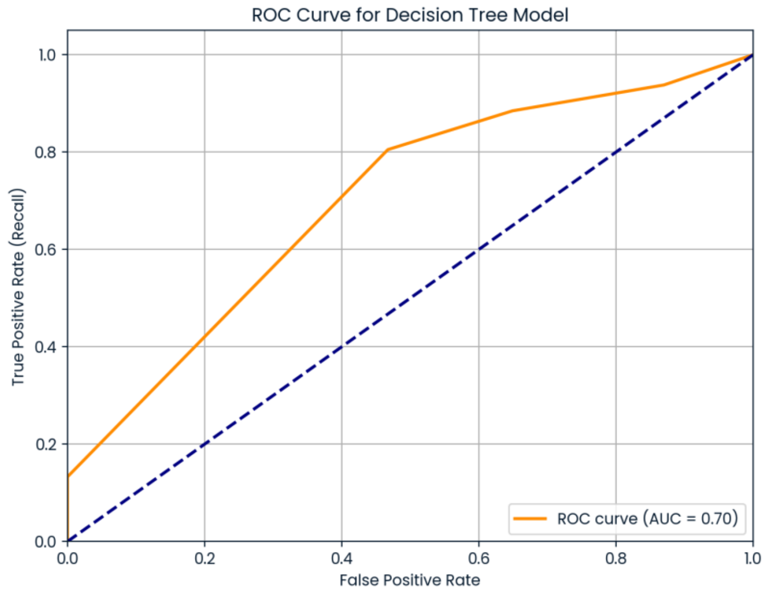
- Recall is a metric used to measure a model's effectiveness at identifying true positives within a specific class.
- In the case of predicting high-traffic recipes, recall calculates the proportion of recipes that were correctly predicted as high traffic out of all actual high-traffic recipes.

Formula:

Recall = True Positives / (True Positives + False Negatives)

Why Recall aligns with the Requirement?

- The requirement: "Correctly predict high traffic recipes 80% of the time" Emphasizes the need to identify most high-traffic recipes.
- Recall directly addresses this need by focusing on capturing as many true high-traffic recipes as possible.
- This approach minimizes missed high-traffic recipes (false negatives), supporting the goal of maximizing website traffic and subscriptions.

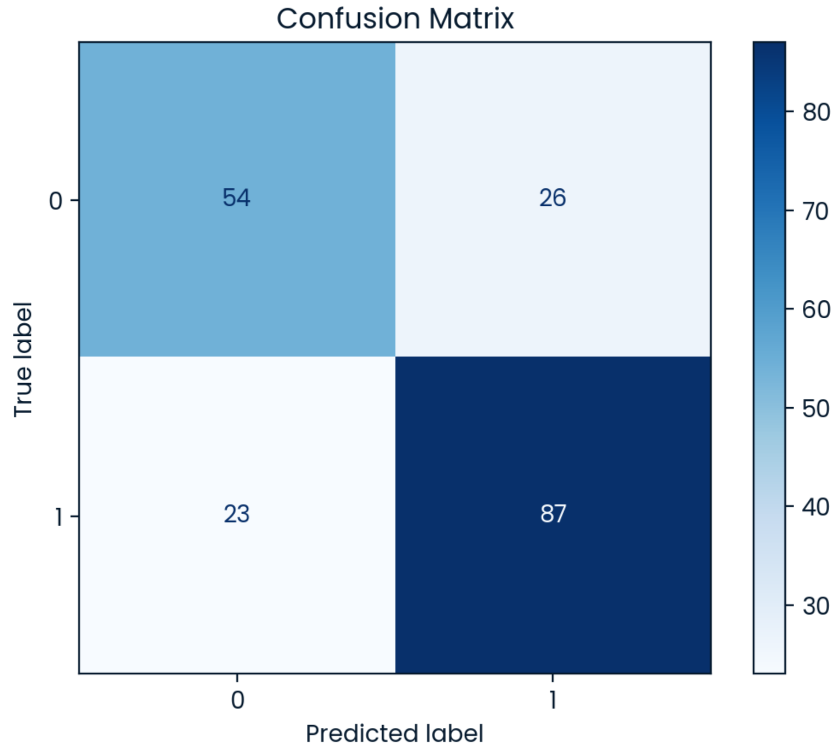


BUSINESS METRIC:
HIGH-TRAFFIC RECIPE COVERAGE RATE (HTRCR)

Definition:
HTRCR is a business-oriented metric that measures the proportion of actual high-traffic recipes
that are successfully predicted and displayed. In essence, it translates the data science metric of
recall into a meaningful business metric.

Formula:
HTRCR = (High-Traffic Recipes Correctly Displayed / Total Actual High-Traffic Recipes) × 100

Why HTRCR aligns with Business Objectives
HTRCR is directly aligned with the business goal of "Correctly predict high-traffic recipes 80% of the time,“
reflecting the model’s capability to maximize engagement by showcasing recipes with high traffic potential.



ALIGNING BUSINESS GOALS WITH THE HIGH-TRAFFIC RECIPE COVERAGE RATE (HTRCR) KPI

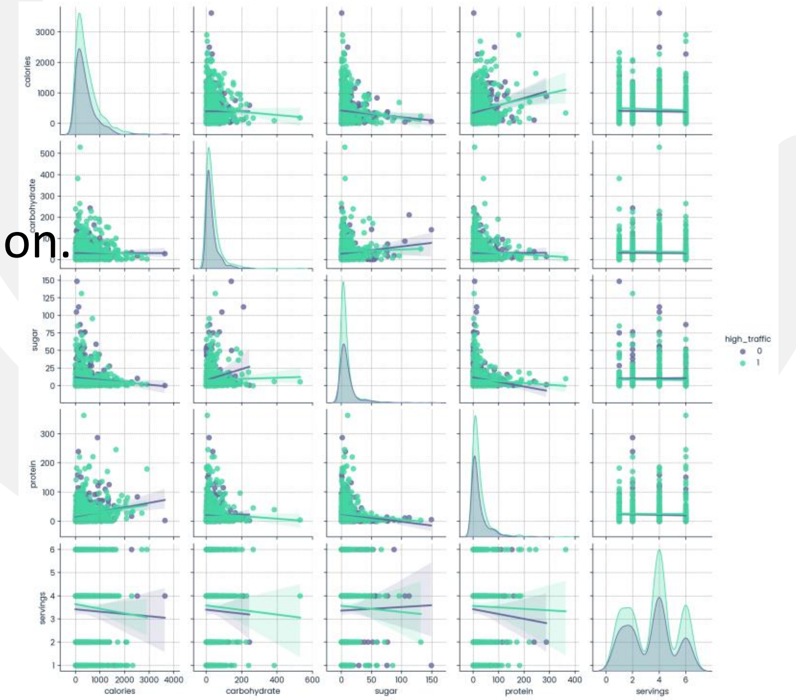
- **Goal: Increase Web Traffic and Engagement**
  Metric Tie-In: A high HTRCR indicates that the model successfully identifies recipes likely to draw attention, potentially driving a significant increase in site visits.

- **Goal: Increase Subscription Rate**
  Metric Tie-In: Tracking HTRCR and analyzing user engagement can help identify recipes that not only drive traffic but also align with conversion efforts, turning visitors into subscribers.

- **Goal: Increase Revenue from Ad Clicks/Views**
  Metric Tie-In: A high HTRCR helps identify recipes likely to perform well in ad-supported sections, justifying strategic ad placements.

- **Goal: Enhance Recipe and Content Strategy**
  Metric Tie-In: A high HTRCR provides insight into user tastes, enabling the marketing and content teams to focus on recipes and categories that resonate with users, enhancing overall engagement.

BUSINESS RECOMMENDATIONS



- **Rotate High-Traffic Recipes:**
  on the homepage based on the predictions of the Final Model in production.
- **Feature Top Probability Recipes:**
  Within the high-traffic predictions, prioritize recipes with the highest predicted probabilities.
- **Monitor and Refine:**
  Tracking the actual traffic generated by these recipes and compare it to model predictions.
- **Content Optimization:**
  Using insights from high-traffic recipes to inform future recipe development.
- **A/B Testing for Engagement:**
  Conducting A/B tests by showing a mix of high- and low-probability recipes on the homepage.
- **High-Traffic Recipe Coverage Rate as KPI:**
  Tracking the High-Traffic Recipe Coverage Rate as a KPI to measure the proportion of actual high-traffic recipes that are accurately identified and featured.
- **Incremental Traffic Tracking:**
  Measuring incremental traffic generated by model-predicted high-traffic.

THANK YOU!